CATCH: Counterfactual Anatomical Tissue Inpainting with Conditional Haar Diffusion

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Abstract. BraTS local synthesis requires generating plausible tumor-free tissue within a masked T1-weighted brain MRI while preserving observed anatomy. We present CATCH, a conditional 3D diffusion framework operating on an invertible Haar-wavelet representation. The denoiser receives noisy target coefficients, voided-image coefficients, and a signed mask. Training combines tumor-excluded wavelet reconstruction with a hole-focused loss, while hard compositing preserves the observed image. We compare fixed-mask training, tumor-component random augmentation, and a weighted mixture of tumor-derived, irregular-blob, and ellipsoidal masks. Within a 25-case development cohort, five prespecified cases select each arm's checkpoint and all 25 select its trajectory-aggregation policy. The frozen pipelines are compared on a separate 75-case internal model-selection set, which selects weighted mixture for organizer evaluation. The selected weighted-mixture pipeline, averaging five stochastic trajectories, achieved SSIM 0.80±0.13, PSNR 19.18±1.80 dB, and MSE 0.010±0.005 (mean±SD). Only this selected pipeline was evaluated on the official 219-case BraTS 2026 validation set, where it achieved SSIM 0.772±0.119, PSNR 20.89±3.27 dB, and MSE 0.0098±0.0054. Relative to compute-matched random augmentation on the internal model-selection set, it improved SSIM by 0.019 (95% bootstrap CI: 0.013–0.025), PSNR by 0.95 dB, and MSE by 0.003; all three paired comparisons remained significant after Holm correction. These results favor the complete weighted-mixture policy within CATCH, while the absence of official fixed- and random-pipeline scores and of a directly comparable external-method baseline limits broader conclusions.

Keywords: Brain MRI inpainting · Healthy-tissue synthesis · Denoising diffusion models · Wavelet transform · Medical image synthesis · BraTS

1 Introduction

Brain tumors alter local anatomy and can undermine registration, parcellation, tissue-segmentation, and atlas-based pipelines developed for non-pathological brains [1]. Healthy-tissue inpainting replaces a pathological region with plausible tumor-free anatomy so that downstream tools can operate on a completed image.

The BraTS local-synthesis task [2] formalizes this setting: given a voided T1-native (T1n) volume and an inpainting mask, synthesize the missing anatomy while leaving the observed image unchanged.

Denoising diffusion probabilistic models (DDPMs) [3] provide a flexible framework for conditional image generation [4,5], but direct full-resolution 3D diffusion is computationally demanding. Wavelet diffusion models (WDMs) [6] reduce the spatial cost via a fixed invertible transform, and conditional WDM [7] extends this idea to paired volumetric synthesis. We adapt this framework as CATCH and study how artificial healthy-tissue holes should be sampled during training.

Training only on supplied masks offers limited spatial and morphological variation. Additional holes may improve coverage, but their source, size, shape, placement, and refresh strategy can alter the learned reconstruction distribution. We therefore compare three mask policies within the same conditional 3D wavelet-diffusion framework. Our main contributions are: (1) a controlled comparison of fixed masks, tumor-component augmentation, and a weighted mixture of tumor-derived, irregular blob, and ellipsoidal masks, including a compute-matched comparison of the two augmented pipelines and development-selected trajectory aggregation for stabilizing stochastic predictions; (2) a full-volume conditional 3D Haar-wavelet diffusion model with tumor-excluded supervision and hard preservation of observed voxels, adapted to support this comparison; and (3) staged selection comprising within-arm checkpoint selection on five prespecified development cases and inference-policy selection across all 25 development cases, followed by frozen 75-case training-policy selection with paired uncertainty, statistical testing, patient-disjoint sensitivity analysis, and qualitative inspection, then blind organizer evaluation of the selected pipeline.

2 Related Work

Diffusion and inpainting. DDPMs learn the reverse of a gradual noising process [3], while score-based formulations connect diffusion to stochastic differential equations [8]. RePaint conditions an unconditional model by repeatedly restoring known pixels during sampling [4], whereas Palette directly supplies the observed image to a conditional denoiser [5]. CATCH follows the latter strategy and applies final hard compositing.

Volumetric medical synthesis. Latent diffusion reduces high-resolution cost through a learned latent space [9]. WDM instead applies an exactly invertible 3D wavelet transform [6], and its conditional extension supports volumetric image-to-image synthesis [7]. Diffusion has also been used for 3D healthy-brain inpainting [10] in the BraTS local-synthesis setting [2], based on the broader BraTS benchmark [11,12,13]. fastWDM3D further reduces wavelet-domain sampling cost through a shortened reverse process [14]. Our work retains full ancestral sampling and focuses on mask representation, tumor-excluded supervision, online healthy-hole sampling, and trajectory aggregation. Differences in data and evaluation preclude direct numerical comparison with prior scores.

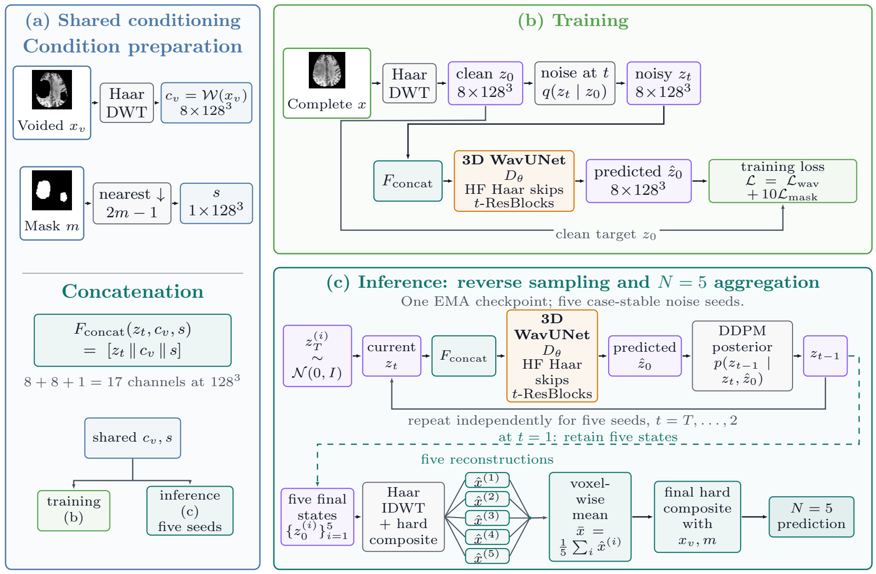

Fig. 1: CATCH training and inference. A one-level 3D Haar transform produces eight subbands. Noisy target coefficients, voided-image coefficients, and the signed mask form the 17-channel WavUNet input. Predictions are inverse-transformed and hard-composited with the observed image. Random and weighted-mixture pipelines average five case-stable noise trajectories from one exponential-moving-average (EMA) checkpoint; fixed mask uses $N = 1$.

3 Method

CATCH adapts Tooth-Diffusion [15] from dental CBCT synthesis to brain-MRI inpainting. Let x_v denote the voided T1n volume and $m \in \{0,1\}^{H\times W\times D}$ the complete inpainting region. During training, m is decomposed into disjoint healthy and pathological components, $m = m_h \vee m_u$ and $m_h \odot m_u = 0$. The complete T1n image x provides a valid target in m_h but remains pathological in m_u, which is therefore excluded from supervision. At inference, only x_v and m are available to the model.

3.1 Conditional Wavelet Diffusion

Volumes are reoriented to RAS and center-padded from $240 \times 240 \times 155$ to 256^3 without resampling. A one-level 3D Haar transform $\mathcal{W}$ produces eight 128^3 subbands. Following WDM [6], the LLL channel is divided by 3 after the forward transform and restored before inversion for target and conditioning coefficients. The transform is therefore fixed and invertible rather than a learned autoencoder.

Let $z_0 = \mathcal{W}(x)$ denote the clean target coefficients. With $T = 1000$, $\alpha_t = 1 - \beta_t$, and $\bar{\alpha}_t = \prod_{j=1}^{t} \alpha_j$, the forward process is

$$q(z_t \mid z_0) = \mathcal{N}\big(z_t; \sqrt{\bar{\alpha}_t}\, z_0, (1 - \bar{\alpha}_t)\mathbf{I}\big)\,, \tag{1}$$

where $\{\beta_t\}_{t=1}^{T}$ discretizes the variance-preserving SDE schedule [8] using the continuous rate parameters $\beta_{\min} = 0.1$ and $\beta_{\max} = 20$. The network predicts $\hat{z}_0$ directly; ancestral sampling uses the DDPM posterior parameterized by $(z_t, \hat{z}_0)$ with the `ModelVarType.FIXED_LARGE` reverse-variance option.

The voided image is transformed as $c_v = \mathcal{W}(x_v)$. The mask is nearest-neighbor downsampled and encoded as $s = 2\,\mathrm{NN}(m) - 1 \in \{-1, 1\}^{128^3}$. The denoiser therefore receives

$$\hat{z}_0 = D_\theta([z_t \| c_v \| s], t), \tag{2}$$

a 17-channel input. D_θ is a fully 3D wavelet U-Net [16,17,6], referred to as *WavUNet* here, with base width 64, channel multipliers $(1, 2, 2, 4, 4, 4)$, two residual blocks per level, GroupNorm, SiLU, timestep-conditioned scale-and-shift modulation, wavelet skip connections, and no self-attention.

3.2 Healthy-Tissue Objective

Target and conditioning image share the inference-available scale $a = \max(x_v)$. Intensities are divided by a, clipped to $[0, 1]$, and mapped to $[-1, 1]$. Let $e_{k,p} = (\hat{z}_0^{(k)}(p) - z_0^{(k)}(p))^2$ denote the error in subband k at wavelet location $p \in \Omega$. We set $u_p = 0$ if the corresponding 2^3 voxel cell intersects m_u, and $u_p = 1$ otherwise:

$$\mathcal{L}_{\text{wav}} = \frac{1}{8|\Omega|} \sum_{k=1}^{8} \sum_{p \in \Omega} u_p e_{k,p}. \tag{3}$$

This matches the implemented fixed-grid normalization: cells touching the real tumor contribute zero error but remain part of the fixed denominator. Because the denominator does not shrink with the excluded region, $\mathcal{L}_{\text{wav}}$ implicitly down-weights the per-voxel training signal for cases with larger tumors, unlike the support-normalized $\mathcal{L}_{\text{mask}}$ below.

For the hole-focused term, cells intersecting m_h form a binary coverage map that is Gaussian-smoothed ($\sigma = 2$ on the wavelet grid) and multiplied by u to obtain $w_p \geq 0$:

$$\mathcal{L}_{\text{mask}} = \frac{\sum_{k=1}^{8} \sum_{p \in \Omega} w_p e_{k,p}}{8 \sum_{p \in \Omega} w_p}, \quad \mathcal{L} = \mathcal{L}_{\text{wav}} + 10\mathcal{L}_{\text{mask}}. \tag{4}$$

When $\sum_p w_p > 0$, the implementation uses Eq. (4); if this support is zero, it assigns that example's regional term zero rather than adding an epsilon. Both terms exclude cells touching the real tumor. Both terms use coefficient-space

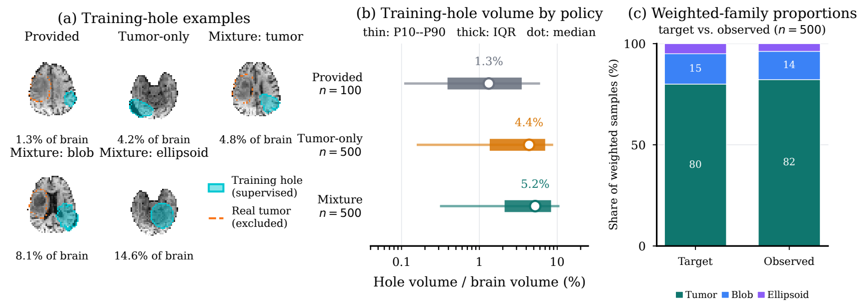

Fig. 2: Training-mask audit. (a) Supplied and generated masks; cyan and orange denote supervised voxels and excluded tumor. (b) Epoch-0 hole-volume distributions (10th–90th percentile, interquartile range, and median; $n = 100/500/500$). (c) Target and observed weighted-mixture proportions: tumor/blob/ellipsoid 80/15/5% and 82.2/14.0/3.8% (411/70/19); no fallback occurred.

mean-squared error, consistent with the direct clean-coefficient regression objective in the adapted wavelet-diffusion implementation. We held this objective fixed across arms to isolate the healthy-mask policy. We did not evaluate an image-space SSIM or perceptual penalty; adding one would change the optimization target and is left to future work.

3.3 Healthy-Mask Training Variants

The *fixed-mask* variant uses the supplied m_h. Online variants sample m'_h, void $m'_h \cup m_u$, and supervise m'_h while excluding m_u. Their tumor-component bank was built from connected components of the corresponding raw BraTS-GLI 2023 tumor-segmentation labels, rather than the supplied enlarged m_u masks; it contains only the 1,151 optimization cases, with five repeat slots per case.

Random augmentation. This reference-style policy uses tumor components only. Current-case components are allowed; source masks are sampled by inverse tumor-to-brain-ratio rank within a conditioned 10% window, transformed by the reference flips and rotations, and fixed across epoch runs. Exhausted placement raises an error rather than silently changing the policy.

Weighted-mixture augmentation. This policy samples 80% tumor components, 15% irregular blobs, and 5% ellipsoids. Current-case tumor components are excluded; empirical target volumes, bounded 3D rotations, optional anisotropic scaling, and epoch-wise slot refresh are used. Accepted masks are connected, contain 800–225,070 voxels, remain at least five voxels from m_u, and overlap background by at most 25%. Exhausted generation falls back to a nonempty supplied healthy mask that does not overlap m_u.

3.4 Sampling and Compositing

We use exponential-moving-average (EMA) weights and the full 1000-step ancestral reverse process. At each step, `clip_denoised=True` inverse-transforms the predicted clean state, clips it in normalized voxel space, and returns it to the wavelet domain. The final reconstruction is restored to native scale, shape, and orientation and hard-composited:

$$\hat{x} = x_v \odot (1 - m) + \hat{x}_{\text{gen}} \odot m. \tag{5}$$

So, outside the hole, the prediction is explicitly copied from the voided input.

Global seed 0 and the case- and member-stable `brats-validation-noise-v1` scheme make trajectories shard-independent and preserve the first members across larger N. Multi-trajectory policies independently decode and composite each sample, aggregate volumes voxel-wise, and composite once more. Thus, trajectories come from one checkpoint rather than independently trained models.

4 Experiments

4.1 Data and Split Protocol

The ASNR-MICCAI BraTS Local Synthesis release [2,11,12,13,18,19,20] contains 1,251 training cases with complete and voided T1n volumes, a complete inpainting mask, and healthy and unhealthy submasks. Organizer-validation ground truth was unavailable during model selection and submission.

A deterministic 100-case hold-out (seed 2026) leaves 1,151 optimization cases, excluded from the hold-out loader and component bank. The hold-out is split into 25 development cases for within-arm checkpoint and inference-policy selection and 75 internal model-selection cases for comparing the frozen training arms. Patients do not overlap between these two subsets, but the split is case-keyed: 3/25 development and 11/75 model-selection cases have sibling timepoints in the optimization pool. None of the five checkpoint-diagnostic cases has such a sibling. We therefore also report a post-hoc analysis on the 64 model-selection cases that are patient-disjoint from optimization.

4.2 Training Configuration

The three 17-channel models differ only in healthy-mask policy. Each uses seed 0, batch size one per GPU, AdamW [21] with learning rate 10^{-5} and zero weight decay, linear decay over a 300,000-step horizon, and EMA decay 0.9999. Training is FP32 on NVIDIA A100 40 GB GPUs. Periodic checkpoints are saved every 5,000 steps through 295,000, followed by a separate terminal save at step 299,999 that includes the EMA state.

4.3 Staged Checkpoint, Inference, and Training-Policy Selection

Within-arm checkpoint and inference-policy selection uses only the 25-case development set and consists of two stages.

Checkpoint selection. Every 10,000 steps, one deterministic 1000-step trajectory is evaluated on five prespecified development cases; fixed-mask step 200,000 is unavailable. Within each training arm, checkpoints are ranked by aggregate SSIM, PSNR, and MSE, and the lowest mean rank is retained. Fixed-mask steps 170,000 and 180,000 tie; step 180,000 is chosen because it leads in PSNR and MSE. Random-augmentation step 290,000 and weighted-mixture step 200,000 uniquely minimize mean rank. These five-case diagnostics are used only for checkpoint selection and are not reported as final performance estimates.

Trajectory-policy selection. At each terminal EMA state, all 25 development cases compare nested $N \in \{1, 3, 5\}$ trajectories with voxel-wise mean or median aggregation. Mean $N = 5$ is selected for both augmented arms. For fixed mask, mean $N = 3$ lowers joint rank from 2.920 to 2.787 but triples inference cost, while mean $N = 5$ is worse (3.240); we therefore retain $N = 1$. Each selected policy is then transferred to the corresponding selected checkpoint and frozen before the 75-case training-policy comparison. This design reduces development-time computation but assumes that the preferred aggregation policy is sufficiently stable across late checkpoints.

Training-policy selection. The three frozen arm-specific pipelines are evaluated once on the separate 75-case set. The pipeline with the lowest mean joint rank across SSIM, PSNR, and MSE is selected for organizer validation; no checkpoint or inference setting is retuned on these cases. Thus, the 75 cases provide an internal model-selection comparison rather than an independent post-selection test.

4.4 Evaluation and Statistical Analysis

For internal evaluation on released training cases, the official evaluator normalizes prediction and target to $[0, 1]$ using the voided-image 0.5th and 99.5th percentiles and scores only the supplied healthy submask. We report SSIM, PSNR, and MSE. Means and sample standard deviations describe case heterogeneity, not training-seed variability.

For case i and pipeline j, let r_{ij}^{SSIM}, r_{ij}^{PSNR}, and r_{ij}^{MSE} denote ranks among the frozen pipelines. Their mean joint rank is $R_j = (3n)^{-1} \sum_{i=1}^{n} (r_{ij}^{\mathrm{SSIM}} + r_{ij}^{\mathrm{PSNR}} + r_{ij}^{\mathrm{MSE}})$, with $n = 75$ for internal training-policy selection; development configurations are ranked analogously within each training arm. For each metric, a Friedman test assesses the omnibus difference and two-sided paired Wilcoxon tests assess the three pairwise contrasts, with Holm correction within metric. The SciPy calls use `zero_method=wilcox`, the asymptotic approximation, and no continuity correction; no reported paired difference is zero. Mean paired differences use deterministic 100,000-resample bootstrap 95% CIs (seed 2026). Predictions, per-case metrics, split audits, hashes, and run metadata are retained in CSV/JSON format. Code and scripts are available at `https://github.com/simonwinther/brats2026`.

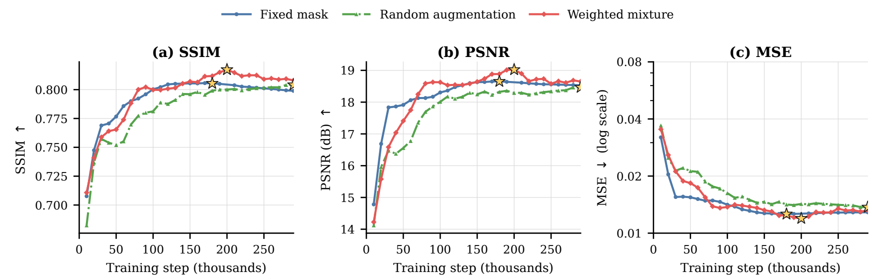

Fig. 3: Five-case EMA checkpoint diagnostics. Stars mark retained checkpoints after ranking aggregate SSIM, PSNR, and MSE. Fixed-mask steps 170k and 180k tie in mean rank; step 180k is retained because it leads in PSNR and MSE. Each point uses one deterministic 1000-step trajectory; curves are within-arm selection diagnostics, not 75-case model-selection results.

Table 1: Development-set joint rank at the terminal EMA states. Bold and underlining mark the lowest and second-lowest values within each arm. Fixed-mask $N = 1$ was retained for the compute–performance trade-off.

Configuration	Fixed mask	Random aug.	Weighted mixture
$N = 1$	2.920	4.707	4.400
Mean $N = 3$	**2.787**	2.133	2.693
Median $N = 3$	2.973	3.533	3.773
Mean $N = 5$	3.240	**1.600**	**1.587**
Median $N = 5$	3.080	3.027	2.547

5 Results

5.1 Development Selection

Figure 3 shows the 5-case diagnostics and selected EMA checkpoints. Fixed-mask steps 170,000 and 180,000 tie for the lowest mean rank; step 180,000 is retained because it leads in PSNR and MSE. Random step 290,000 and weighted-mixture step 200,000 are best on all aggregate metrics within their respective histories.

Table 1 summarizes trajectory-policy selection on all 25 development cases. Mean $N = 5$ has the lowest joint rank for both augmented models. For fixed mask, mean $N = 3$ improves rank by only 0.133 over $N = 1$ at triple trajectory cost, and mean $N = 5$ is worse. We therefore select fixed mask with $N = 1$ and both augmented models with mean $N = 5$.

5.2 Internal 75-Case Training-Policy Selection

Table 2 summarizes the frozen pipelines on the internal model-selection set. Omnibus differences are significant for SSIM ($\chi^2_F(2) = 52.83$), PSNR, and MSE

Table 2: Internal 75-case training-policy selection results. Values are mean ± sample SD. Joint rank averages each pipeline's per-case SSIM, PSNR, and MSE ranks; lower is better. Bold and underlining mark the best and second-best value in each numeric column. Fixed, Random, and Weighted denote the fixed-mask, random-augmentation, and weighted-mixture pipelines. Fixed uses $N = 1$, so its contrasts with the mean-$N = 5$ pipelines are not compute matched.

Pipeline	SSIM ↑	PSNR (dB) ↑	MSE ↓ ($\times 10^{-2}$)	Rank ↓
Fixed ($N = 1$)	0.7561 ± 0.1615	17.84 ± 1.75	1.3645 ± 0.8066	2.551
Random (mean $N = 5$)	<u>0.7784 ± 0.1398</u>	<u>18.24 ± 2.19</u>	<u>1.2515 ± 0.7176</u>	<u>2.076</u>
Weighted (mean $N = 5$)	**0.7972 ± 0.1308**	**19.18 ± 1.80**	**0.9874 ± 0.5359**	**1.373**

(both $\chi^2_F(2) = 52.91$; all $p < 3.4 \times 10^{-12}$). Relative to fixed mask, weighted mixture improves SSIM by 0.041 (95% CI: 0.032–0.050), PSNR by 1.34 dB (1.08–1.61), and $[0, 1]$-scale MSE by 0.0038 (0.0029–0.0047). Weighted mixture leads all three metrics and has the lowest joint rank, so it is selected for the organizer evaluation. This arm-specific-policy comparison is not compute-matched; an additional all-$N = 5$ sensitivity analysis below isolates the effect of inference trajectory count.

Relative to compute-matched random augmentation, weighted mixture improves SSIM by 0.019 (0.013–0.025), PSNR by 0.95 dB (0.66–1.27), and $[0, 1]$-scale MSE by 0.0026 (0.0019–0.0034). All three pairwise tests within each metric remain significant after Holm correction; the largest adjusted p-value is 0.0178.

As a post-hoc check separating inference compute from training-policy effects, Table 3 reports an additional analysis on the same internal model-selection set in which the fixed-mask checkpoint uses the same five case-stable trajectories and voxel-wise mean aggregation as the augmented pipelines. Fixed mean-$N = 5$ achieves SSIM 0.7561 ± 0.1615, PSNR 17.845 ± 1.750 dB, and MSE 0.01364 ± 0.00806. Its means differ from the fixed-$N = 1$ result by only -0.000005 SSIM, $+0.0006$ dB PSNR, and -0.000002 MSE. With all pipelines compute-matched at mean $N = 5$, weighted mixture exceeds fixed mask by 0.0411 SSIM (95% CI: 0.0324–0.0500), 1.339 dB PSNR (1.077–1.608), and an MSE reduction of 0.00377 (0.00288–0.00474), winning 66/75 cases on each metric. These three weighted-versus-fixed contrasts have Holm-adjusted $p < 2.5 \times 10^{-11}$, and all nine pairwise metric contrasts remain significant (largest adjusted $p = 0.0180$). Thus unequal trajectory count does not account for selecting weighted mixture over fixed mask.

Figure 4 shows that the compute-matched gain occurs in most cases. Weighted mixture outperforms random augmentation in 54/75 cases for SSIM and 57/75 for PSNR and MSE. It improves all three metrics in 51 cases, loses all three in 15, and has mixed effects in 9. Across all pipelines, it is simultaneously best on all three metrics in 46 cases and attains the lowest mean joint rank.

In a post-hoc exploratory analysis, we stratified weighted-minus-random differences by tertiles of the provided healthy scoring-mask volume ($n = 25$ each),

Table 3: Post-hoc compute-matched sensitivity analysis on the 75-case internal model-selection set. Values are mean ± sample SD and all pipelines use voxel-wise mean aggregation with $N = 5$. Joint rank averages each pipeline's per-case SSIM, PSNR, and MSE ranks; lower is better. Bold and underlining mark the best and second-best value in each numeric column. This sensitivity does not replace the frozen, development-selected comparison, in which fixed mask uses $N = 1$.

Pipeline	SSIM ↑	PSNR (dB) ↑	MSE ↓ ($\times 10^{-2}$)	Rank ↓
Fixed (mean $N = 5$)	0.7561 ± 0.1615	17.85 ± 1.75	1.3642 ± 0.8065	2.551
Random (mean $N = 5$)	<u>0.7784 ± 0.1398</u>	<u>18.24 ± 2.19</u>	<u>1.2515 ± 0.7176</u>	<u>2.076</u>
Weighted (mean $N = 5$)	**0.7972 ± 0.1308**	**19.18 ± 1.80**	**0.9874 ± 0.5359**	**1.373**

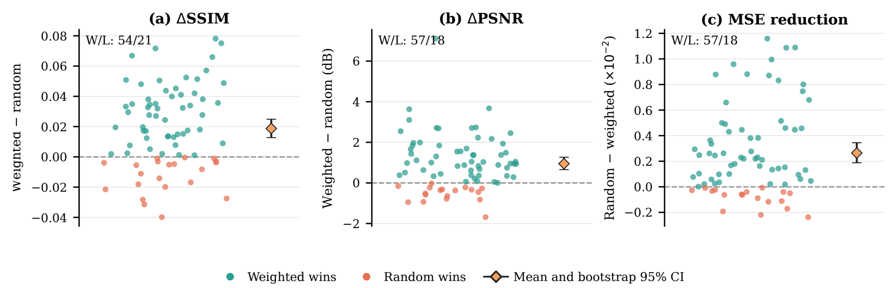

Fig. 4: Compute-matched paired effects of weighted mixture versus random augmentation across the 75 internal model-selection cases. Positive values favor weighted mixture. Points represent cases; diamonds show mean paired effects with 100,000-resample case-bootstrap 95% CIs. W/L denotes wins/losses.

explicitly excluding the complete inpainting mask from the size definition. For small, medium, and large scored holes, respectively, ΔSSIM was +0.012, +0.023, and +0.022; ΔPSNR was +1.08, +0.88, and +0.88 dB; and ΔMSE was −0.0020, −0.0031, and −0.0028. SSIM and MSE gains were smaller for small holes; we do not interpret medium–large differences.

Figure 5 shows variation in case difficulty. The low-SSIM example preserves gross structure but shows localized intensity and boundary mismatch; the median retains residual error; and the high-SSIM example closely matches the observed target. Because the observed T1n may remain pathological elsewhere in the full inpainting region, these scores do not establish counterfactual accuracy there.

The weighted-mixture mean-$N = 5$ pipeline selected on the internal 75-case set was the only pipeline submitted to the official BraTS 2026 evaluator. Its exact archive (Synapse submission 9774333) was scored on 219 validation cases and achieved SSIM 0.7723 ± 0.1192, PSNR 20.887 ± 3.273 dB, and MSE 0.009824 ± 0.005448. Fixed and random were not submitted, so official 219-case scores are unavailable for those pipelines and the organizer result is not a

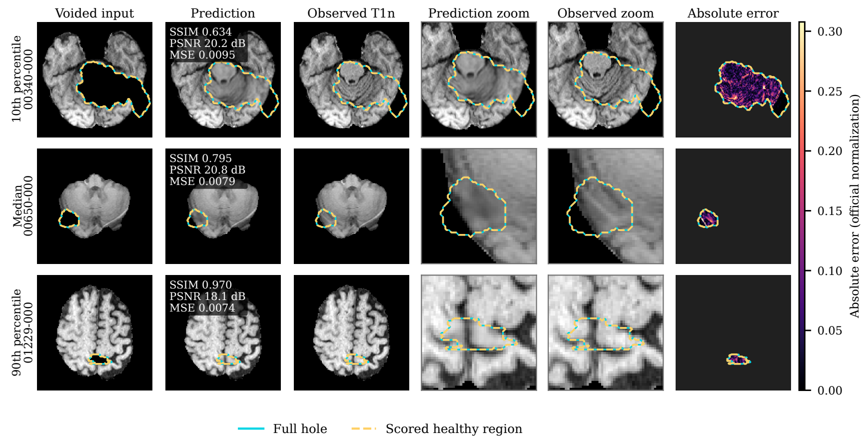

Fig. 5: Weighted-mixture mean-$N = 5$ reconstructions at the empirical 10th, 50th, and 90th SSIM percentiles. Cyan denotes the complete inpainting region and dashed yellow the scored healthy region. Error is shown only within the scored region on a shared clipped scale.

three-arm comparison. Table 4 places this blind weighted-pipeline result beside reported BraTS 2023 test-set results for context. The challenge year, cohort, evaluation split, and methods differ, so the rows are not a controlled comparison and do not establish relative rank [14].

6 Discussion

Weighted mixture is the strongest pipeline on the 75-case internal model-selection set and is therefore selected for organizer evaluation. Random and weighted mixture use the same architecture, training horizon, mean-$N = 5$ inference, split, noise trajectories, and evaluator, so their comparison is compute-matched. Their training policies nevertheless differ jointly in mask families, volume sampling, transforms, current-case eligibility, epoch refresh, and fallback. The evidence therefore supports selecting the complete policy rather than its 80/15/5 proportions or any single component. Improvements in most cases argue against an outlier-only effect, but 15 all-metric losses show that the method is not uniformly superior. Because these 75 cases determine the selected arm, their estimates and hypothesis tests describe the internal selection comparison rather than independent post-selection confirmation. One training seed and arm-specific checkpoints limit attribution.

The frozen, development-selected fixed-mask pipeline uses one trajectory. Fixed mask uses $N = 1$ because mean $N = 3$ provided only a small development-rank gain at triple cost and mean $N = 5$ was worse. Its comparisons with the augmented pipelines therefore combine training-policy and inference-compute

Table 4: BraTS 2023 test-set results reported by Durrer et al. [14] and the blind CATCH BraTS 2026 validation result (mean ± SD). The first three rows are the 2023 challenge podium methods; fastWDM3D and plain WDM3D are later reported evaluations. Only the CATCH row uses mean-$N = 5$ aggregation. Challenge years, cohorts, and evaluation splits differ, so rows are not directly comparable.

Method	SSIM ↑	PSNR (dB) ↑	MSE ↓
Zhang et al. (1st, BraTS 2023)	0.91 ± 0.15	23.59 ± 5.35	0.0049 ± 0.0016
Durrer et al. (2nd, BraTS 2023)	0.86 ± 0.20	20.42 ± 3.82	0.0100 ± 0.0016
Huo et al. (3rd, BraTS 2023)	0.87 ± 0.18	18.71 ± 4.01	0.0144 ± 0.0025
fastWDM3D ($T = 2$)	0.86 ± 0.12	22.26 ± 3.97	0.0079 ± 0.0063
WDM3D (plain)	0.61 ± 0.16	10.57 ± 3.20	0.1060 ± 0.0757
CATCH, weighted	0.772 ± 0.119	20.89 ± 3.27	0.0098 ± 0.0054

differences. In the post-hoc compute-matched sensitivity analysis that equalizes inference at mean $N = 5$, fixed-mask performance is practically unchanged and weighted mixture retains its 0.0411 SSIM, 1.339 dB PSNR, and 0.00377 MSE advantages. This removes unequal inference compute as an explanation for the observed gap, while the development-selected $N = 1$ result remains the primary fixed-mask policy.

Wavelets enable full-volume 3D sampling, while hard compositing preserves the observed input outside the hole. However, mean-$N = 5$ inference requires five 1000-step trajectories per case. On NVIDIA A100 40 GB GPUs, steady-state wall-clock time was 5.86 minutes per case for $N = 1$ and 28.96–29.11 minutes per case across the random and weighted mean-$N = 5$ runs, or 4.94–4.97 times the $N = 1$ time. These measurements use intervals between consecutive saved predictions within each deterministic shard and exclude one-time model startup. Faster wavelet inpainting [14] should therefore be evaluated under the same cohort and protocol.

Evaluation covers only artificially removed healthy voxels. Since the observed T1n remains pathological in m_u, neither quantitative metrics nor qualitative references establish patient-specific counterfactual correctness in the real tumor region. Plausible reconstructions may still hallucinate incorrect anatomy, and clinical validity remains untested. The reported 219-case values are the organizer-produced blind scores. Local package-integrity checks did not access ground truth or scoring masks, were not used for model selection or arm comparison, and are not additional performance results.

The split is case-keyed rather than fully patient independent: the optimization pool contains sibling timepoints for $3/25$ development and $11/75$ model-selection cases. In the 64 model-selection cases patient-disjoint from optimization, weighted mixture obtains SSIM 0.810, PSNR 19.27 dB, and MSE 0.0092; random obtains 0.791, 18.33 dB, and MSE 0.0117; and fixed mask obtains 0.770, 17.93 dB, and MSE 0.0128. The ordering is unchanged, although development-stage selection may still be influenced by optimization-pool siblings.

6.1 Limitations

Other limitations are one seed per arm, five-case single-trajectory checkpoint selection, which risks selection bias and noise-seed variance that a larger development pool could reduce in future iterations, transfer of terminally selected inference policies to arm-specific checkpoints, and no directly comparable external-method baseline on the same BraTS 2026 cohort. The fixed-mask mean-$N = 5$ experiment and void-size stratification are post-hoc sensitivity analyses; neither replaces the frozen development-selected protocol, in which fixed mask uses $N = 1$. The 75-case comparison selects the training policy as well as quantifying its internal advantage, and only weighted mixture was evaluated on the official 219-case cohort; consequently, the official result cannot independently confirm the fixed–random–weighted ordering. Across-case SDs and bootstrap CIs do not measure training-seed variability, so superiority to prior methods is unestablished. Development predictions and the exact dirty-worktree differences for the random arm and post-hoc fixed-mask mean-$N = 5$ run were not preserved, although configurations, metrics, metadata, prediction hashes, and final model-selection outputs remain archived. Future work should add seeds and baselines, evaluate faster samplers, adopt patient-level splits, and investigate multimodal conditioning and anatomy-aware plausibility.

7 Conclusion

We presented CATCH, a conditional 3D Haar-wavelet diffusion framework for healthy brain-tissue inpainting. The method combines full-volume wavelet-domain diffusion, tumor-excluded reconstruction supervision, hole-focused loss weighting, and hard compositing of the observed image. Within CATCH, weighted-mixture augmentation outperformed compute-matched tumor-component augmentation on the 75-case internal model-selection set, with the same ordering in the patient-disjoint sensitivity analysis. This comparison selected weighted-mixture mean-$N = 5$ for submission; the selected pipeline achieved SSIM 0.772, PSNR 20.89 dB, and MSE 0.0098 on the official 219-case validation set. An additional all-$N = 5$ sensitivity analysis preserved the weighted-versus-fixed ordering, showing that unequal inference trajectory count did not drive the internal selection. Because only weighted mixture received official evaluation, the 219-case result evaluates the selected pipeline but does not compare the three arms. These findings support the complete weighted-mixture policy under the reported configuration, but the absence of a directly comparable external-method baseline and evaluation only on artificially removed healthy tissue limit broader claims.

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